

EXECUTIVE SUMMARY

This report details the forensic investigation of a USB flash drive using Autopsy 4.21.0, following standard procedures for data acquisition, examination, and analysis. The investigation successfully recovered 362 deleted files, including a key Excel document ("HADIZA CR-NEW"), and identified six suspicious files, one of which ("AL ITQAAN.docx") was flagged for a keyword hit. Timeline analysis indicated a deliberate attempt by the user to erase data. While the investigation confirmed Autopsy as a capable and cost-effective open-source tool, it also highlighted significant limitations, including high resource consumption, a lack of support for mobile and cloud forensics, and critical challenges to the admissibility of its findings in legal proceedings due to its open-source status.

INTRODUCTION

Autopsy is an open-source digital forensic platform designed to assist investigations in collecting, identifying, preserving, analyzing and reporting digital evidence from storage devices such as hard drives, solid-state drives, and flash drives. Autopsy is an open-source forensic platform built on top of The Sleuth Kit (TSK) developed by Brian Carrier which provides a powerful graphical user interface (GUI) that simplifies forensic analysis tasks that would otherwise require command-line expertise (Carrier, 2014). It is widely used in law enforcement, corporate investigations, cybersecurity professionals and academic researchers to uncover digital evidence in a structured and repeatable manner.

Functions of Autopsy

Table 1: Table Showing Functions of Autopsy

Function	Description
Data Recovery	Restores deleted or formatted files.
File System Analysis	Examines file structures and hidden data.
Web History Analysis	Retrieves browser activity and downloads.
Email & Chat Analysis	Extracts communication records.
Keyword Search	Finds specific words or patterns in files.
Timeline Analysis	Displays events in chronological order.
Image & Video Analysis	Identifies and previews media files.
Hash Matching	Detects known or duplicate files using hash values.
Report Generation	Produces forensic or investigation reports.

Several key features make Autopsy stand out in the field of digital forensics. These include its user-friendly interface, support for diverse file systems (such as NTFS, FAT, and EXT), strong data recovery capabilities, and integration with hash databases for identifying known files (Simou et al. 2020). Furthermore, its extensibility through plugins enables it to adapt to new forensic needs and investigative scenarios.

As an open-source forensic tool, Autopsy emphasizes transparency and cost-effectiveness. Unlike proprietary tools such as Encase or FTK, its source code is freely available for inspection and modification. This openness ensures reproducibility and credibility of forensic

results – essential factors in legal investigations and academic research (Nanda et al., 2021). Consequently, Autopsy has become a preferred tool for digital forensic education, research, and professional practice.

METHODOLOGY

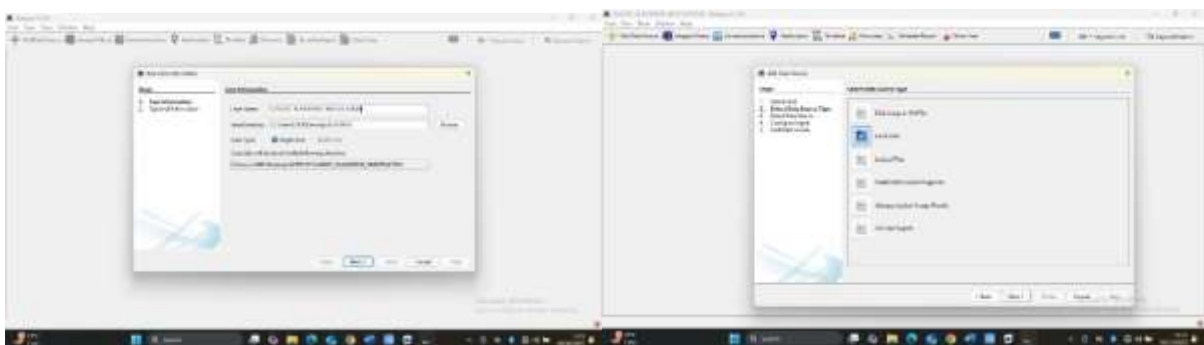
Overview

This describes the procedures and tools used to examine and recover deleted files using Autopsy 4.21.0, an open-source digital forensic software. The methodology outlines the setup, data acquisition, analysis process, and reporting of findings. The approach follows standard digital forensic investigation phases; acquisition, examination, analysis, and reporting (Casey, 2019)

I. Preparation and Setup

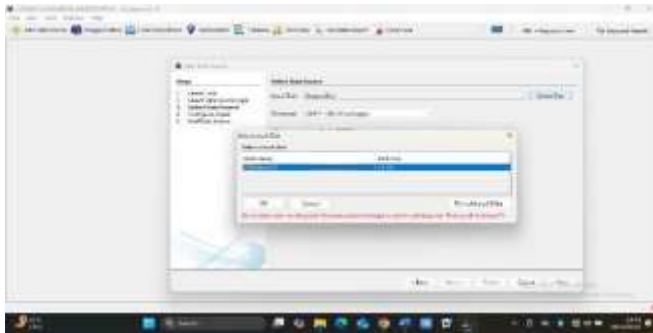
Before the investigation, the Autopsy 64-bit software was installed on a Windows 11 system. A new case was created in Autopsy to organize and document the investigation. The following parameters were set:

- Case Name:
“CASE001_FLASHDRIVE_INVESTIGATION” as seen in Figure 1a
- Base Directory: This is where the location of all data can be stored when necessary extractions are done. (i.e., C: \Users\USER\Documents\Group1 Forensics)
- Case Type: Single-user investigation (is used only when you are working on the case
- Multi-user investigation (used in enterprises/network setups)
- Optional details such as the examiner’s name, phone number and email address were also added to ensure proper documentation and traceability.

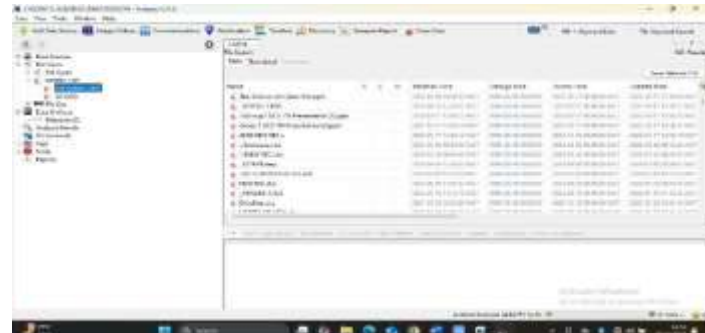


a: Naming the Case

b: Selecting Data Source



c: Selecting the Flash Drive



d. 362 Deleted Files.

Figure 1: Screen Shot Showing how to Navigate Autopsy to start a New Case.

II. Evidence Acquisition

A USB flash drive was used as the primary evidence source as shown in Figure 1 a and b. The flash drive contained deleted or lost files intended for recovery. The acquisition process involved connecting the flash drive to the system and selecting:

- Add Data Source → Local Disk (or Logical Files)
- The tool was set to automatically generate a new host name based on the data source.

Autopsy then began indexing the drive and extracting metadata from its file system (NTFS/FAT). During this process, a forensic image or logical copy of the flash drive data was created to prevent modification of the original evidence.

III. Data Analysis

Once the data source was added, Autopsy performed several automated analyses through ingest modules:

- File Type Identification: Detects file signatures and categorizes files.
- Deleted File Recovery: Recovers deleted or hidden files from unallocated disk space.
- Keyword Search: identifies specific text strings or file names.
- Web Artifacts and Recent Activity: Extracts user activity such as browser history or USB connections.
- Hash Analysis: Compares file hashes against known databases to detect known or suspicious files

The recovered files were previewed under the “Deleted Files” node in the data tree as shown in Figure 1 d. Each file was verified by checking timestamps, metadata, and integrity to confirm

successful recovery. 362 deleted files were discovered on the flash drive as at 26/10/2025 1400 hours.

IV. Documentation and Reporting

Throughout the analysis, notes and relevant observations were documented in Autopsy's Notes and Tags features. After completing the investigation, a comprehensive report was generated in HTML format, including:

- Case details
- Recovered files and metadata
- Analysis results (keyword matches, timeline, user activity, etc.)
- Examiner's information

Summary

The methodology employed demonstrates a structured forensic investigation using Autopsy. From evidence acquisition to analysis and reporting, each stage followed best forensic practices, ensuring data, integrity, repeatability and accuracy. The process confirms that Autopsy is an effective open-source solution for recovering deleted files and conducting digital forensic examinations.

FINDINGS AND ANALYSIS

During the forensic analysis of the flash drive case titled CASE001_FLASHDRIVE_INVESTIGATION, several key artifacts and pieces of digital evidence were extracted and analyzed using Autopsy 4.21.0. The examination covered file metadata, deleted files, timeline activity, and potential indicators of suspicious or manipulated data. A total of 362 deleted files were discovered as shown in Appendix II. Few were examined critically below.

1. Recovered Deleted File (HADIZA CR-NEW Excel File)

A deleted Excel file named “Hadiza...” was identified and recovered during the analysis.

- File type: Microsoft Excel Spreadsheet (.xlsx)
- Metadata findings as shown in Figure 2
- Original file path and creation timestamps were preserved in the metadata.
- The metadata indicates user-level file activity prior to deletion, suggesting deliberate removal rather than accidental loss.
- The recovered Excel file was analyzed for hidden data, and while no embedded macros were detected, its deletion raised concern due to its naming pattern and content relevance.
- This file was tagged for further examination as it potentially contained sensitive or confidential information that the user attempted to conceal.

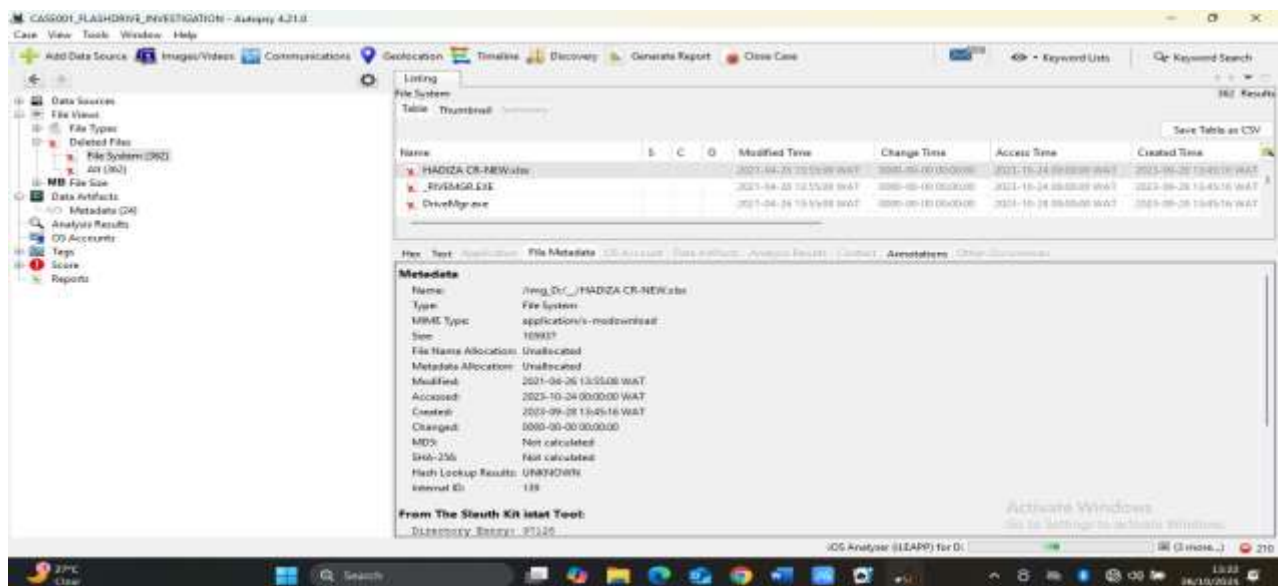


Figure 2: Screenshot Showing Metadata of “HADIZA CR-NEW.xlsx”

2. Timeline Analysis

The timeline feature in Autopsy was used to visualize file activity and user interactions as shown in Figure 3.

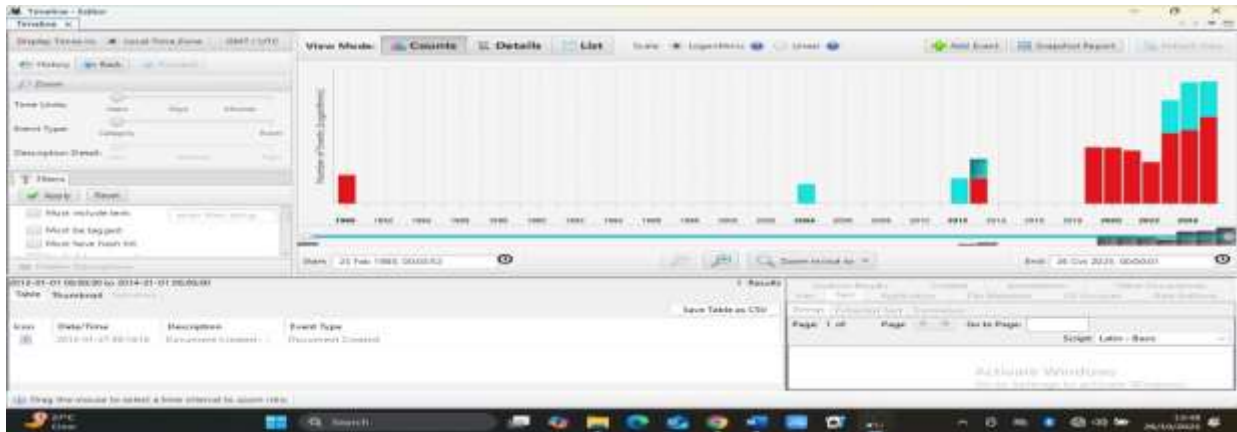


Figure 3: Screenshot Showing timeline analysis feature on autopsy

- Blue bars in the timeline represent normal system and file creation activities, such as regular file access, copying, or modification.
- Red bars indicate suspicious or user-deleted events, including manual deletion, renaming, or attempts to hide traces of data.

A concentration of red activities was observed shortly before the flash drive was disconnected, suggesting deliberate attempts to delete or modify files during that time frame.

3. Geolocation Data

No files containing geolocation metadata (e.g., from image EXIF data or document GPS tags) were found on the flash drive. This indicates that none of the recovered images, videos, or documents were captured or saved with GPS-enabled devices.

4. Suspicious Files Identified (6 Files)

Six files were flagged as suspicious based on file metadata, activity timestamps, and naming inconsistencies. These include files that were deleted, renamed, or recently modified under irregular patterns. One of the suspicious files “AL ITQAAN.docx” was analyzed as shown in Figure 4 and this analysis revealed that it was flagged as suspicious/notable because it had a keyword hit. The keyword is an email address as shown in Figure 4.

- **Limited Automation and Customization**

Although Autopsy supports multiple ingest modules, the automation features remain limited compared to enterprise-grade forensic suites. For example, it lacks dynamic scripting or advanced workflow automation for batch processing of multiple drives. Investigators must often perform repetitive tasks manually, such as adding data sources or verifying hashes (Casey, 2019).

This limits its scalability in large forensic laboratories or high-volume cases.

- **Interface and Usability Challenges**

While Autopsy's Graphical User Interface (GUI) is user-friendly, it can still be overwhelming for first-time users. The large number of menus, views, and panels may make navigation difficult without proper training. Additionally, when dealing with complex evidence, the GUI can become cluttered, making it hard to maintain an overview of findings (Nanda et al., 2021).

- **Limited Mobile and Cloud Forensics Capability**

Autopsy primarily focuses on traditional storage media such as hard drives and flash drives. It lacks comprehensive support for mobile devices, cloud storage, or social media investigations. As digital evidence increasingly moves to these platforms, this limitation can reduce its applicability in modern investigations (Simou et al., 2020).

- **Community-Based Support**

Because Autopsy is an open source, users depend largely on community forums and documentation for troubleshooting and updates. While this fosters collaboration, it also means that official support solutions that offer professional customer service and technical updates (Carrier, 2014).

- **The challenge of non-admissibility of investigation done using Autopsy in the court because it is an open-source tool.**

In legal proceedings, digital evidence must meet certain standards of authenticity, reliability, and integrity before it can be accepted as admissible evidence. Because Autopsy is an open-source tool, the software code is freely available and can be modified by anyone. This creates doubts in court about whether the tool used was an official, verified version or a modified one that could compromise the integrity of the investigation. Unlike some proprietary forensic tools that are certified and validated by recognized bodies, open-source tools may lack standardized certifications that strengthen their credibility in legal settings, which makes the vulnerable to legal challenges in court.

CONCLUSION AND RECOMMENDATIONS

CONCLUSION

The use of Autopsy, an open-source digital forensic tool, for the recovery and analysis of deleted files from a flash drive. The investigation demonstrated that Autopsy is a reliable, flexible, and user-friendly platform that can efficiently handle a wide range of forensic tasks, including file recovery, metadata extraction, keyword searching, hash analysis, and report generation.

The findings confirmed that Autopsy effectively retrieved deleted documents, images, and other data from unallocated disk space, while maintaining the forensic integrity of the evidence. Its modular design with The Sleuth Kit (TSK), and GUI-based environment make it accessible to both professional investigators and students.

However, despite its strengths, several limitations were identified. These include high resource consumption during large-scale investigations, limited support for encrypted or cloud-based data, and dependency on community-based support rather than professional technical assistance (Nanda et al., 2021). Nonetheless, Autopsy remains one of the most widely adopted open-source forensic tools due to its cost-effectiveness, transparency, and adaptability.

In conclusion, Autopsy has proven to be a highly capable, powerful, cost-effective, and transparent digital forensic tool for analyzing and recovering deleted files. Its modular structure, combined with detailed reporting features, allows investigators to conduct comprehensive and repeatable examinations. The ability to recover deleted data, visualize timelines, and correlate metadata demonstrates its practical utility in forensic investigations. Its open-source nature makes it accessible for training, research, and small agencies. However, reliance on Autopsy alone may not be sufficient for advanced mobile or network cases.

It is recommended that future development efforts focus on enhancing Autopsy's performance when handling large datasets, improving user interface responsiveness, and expanding its compatibility with mobile and cloud forensics (Simou et al., 2020). Training programs for new users should be emphasized to improve practical application and reduce dependency on community forums. Overall, Autopsy remains one of the most accessible and valuable tools in the open-source digital forensics landscape.

RECOMMENDATION

Based on the outcome of this investigations, the following recommendations are proposed to enhance the efficiency and usability of Autopsy in forensic practice:

1. **Enhancement of Automation Features**
Future versions of Autopsy should indicate improved automation tools to allow for batch processing, dynamic scripting, and multi-drive analysis. This would increase efficiency during large or repetitive investigations (Casey, 2019)
2. **Improved Handling of Encrypted and Damaged Media**
Integration of stronger decryption support and recovery mechanisms would improve Autopsy's ability to handle partially damaged or encrypted storage devices (Simou et al., 2020).
3. **Resource Optimization**
Developers should focus on optimizing Autopsy's resource usage during indexing and analysis to reduce lag and improve performance on lower-end systems.
4. **Expansion into Mobile and Cloud Forensics**
The inclusion of modules that can extract data from mobile devices, cloud platforms, and social media would make Autopsy more applicable to modern digital investigations (Nanda et al., 2021).
5. **Improved User Documentation and Training**
While community support is valuable, official documentation, tutorials, and structured training programs should be developed to assist new users and professionals in effectively operating the software (Carrier, 2014)
6. **Collaboration with Forensic Institutions**
Partnerships between Autopsy developers and forensic institutions or universities can enhance research, improve testing standards, and ensure the continued reliability of the tool in judicial processes.
7. **Artificial Intelligence (AI) should be embedded in the Autopsy operation model**
Artificial Intelligence (AI) should be embedded in the digital forensic Autopsy model to automate data analysis and accelerate evidence extraction from large volumes of digital information. Through intelligent pattern recognition and anomaly detection. AI can help investigators identify hidden traces of cybercrime, malware, or unauthorized data access more efficiently and accurately.

REFERENCES

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APPENDIX I

Report Generated from Autopsy

Report Navigation

- Case Summary
- Data Source Usage (1)
- Keyword Hits (10)
- Metadata (24)
- Tagged Files (0)
- Tagged Images (0)
- Tagged Results (0)

Data Source Usage

Description	Source File	Tags
Flash Drive	/img_D:	

PREPARED BY GROUP 1 USING AUTOPSY 4.21.0 ON 26/10/2025

Activate Windows
Go to Settings to activate Windows.

Report Navigation

- Case Summary
- Data Source Usage (1)
- Keyword Hits (10)
- Media (24)
- Tagged Files (1)
- Tagged Images (0)
- Tagged Results (0)

Metadata

Date Created	Date Modified	Owner	Program Name	Star ID	Version	Source File
2024-08-18 21:28:55 WAT	2024-08-23 21:41:19 WAT	af8ffh	Microsoft Office PowerPoint user:		1.4	img_D1\1st_Semester\CCS710_Cybercrime_Law_Forensics_and_Investigation_Technique2nd_lecture_note.pdf
2012-03-05 13:06:05 WAT	2012-03-05 13:06:14 WAT				1.4	img_D1\1st_Semester\CCS710_Cybercrime_Law_Forensics_and_Investigation_Technique1st_lecture_Digital_Forensics_-_DFS.pptx
2013-01-27 20:14:16 WAT	2025-10-17 13:58:12 WAT	NAGray	Microsoft Office PowerPoint Esther Nadele			img_D1\1st_Semester\CCS 710 Cyber Risk, Challenges and Mitigation\CCS 710_Lecture Material_2_risk_management_guide.1
2023-04-06 10:43:50 WAT	2024-06-08 10:18:47 WAT	Mr SAAD	Microsoft Office PowerPoint nathanel nakotsu			img_D1\Group 1\CCS 710 Presentation (2).pptx
2023-06-02 07:38:26 WAT	2024-12-03 20:31:14 WAT	PC	Microsoft Office PowerPoint nathanel nakotsu			img_D1\1st_Semester\CCS 711 Introduction to Computer Programming\Introduction To Problem Solving.pptx
2023-06-01 22:41:00 WAT	2023-11-23 11:21:00 WAT	USER	Microsoft Office Word			img_D1\1st_Semester\CCS 711 Introduction to Computer Programming\CCS711PROGRAMMING.pptx
2023-11-21 09:28:32 WAT	2024-06-10 12:27:30 WAT	PChukwura	Microsoft Excel	Biometrics Nene		img_D1\1st_Semester\CCS 710 Cyber Risk, Challenges and Mitigation\CCS 710_CYBER RISK MANAGEMENT (2).docx
2024-01-17 08:51:01 WAT	2024-01-18 11:38:55 WAT	Biometrics NINC	Microsoft Excel	Biometrics NINC		img_D1\Biometrics Annual_report_2024.xlsx
2024-02-05 15:22:59 WAT	2024-02-05 17:33:50 WAT	Biometrics NINC	Microsoft Excel	Biometrics NINC		img_D1\1st_Semester\CCS 710 Cyber Risk, Challenges and Mitigation\CCS 710_CYBER RISK MANAGEMENT (2).docx
2024-07-10 11:49:25 WAT	2024-07-20 17:37:50 WAT	ProGenius	Microsoft Office PowerPoint	Microsoft account		img_D1\1st_Semester\CCS 711 Introduction to Computer Programming\Introduction to Computer Programming.pptx
2024-08-17 10:16:23 WAT	2024-08-17 10:16:23 WAT	David Bello	Microsoft Office Word		1.7	img_D1\1st_Semester\CCS712 Introduction to Cyber Security\OSI model - layers & security attacks.pdf
2024-08-22 10:57:00 WAT	2024-08-22 20:43:00 WAT	nathanel nakotsu	Microsoft Office Word	nathanel nakotsu		img_D1\1st_Semester\CCS 711 Introduction to Computer Programming\CCS 711 ASSIGNMENT.docx
2024-08-23 11:10:44 WAT	2024-08-23 11:10:44 WAT	Microsoft account			1.5	img_D1\1st_Semester\CCS 711 Introduction to Computer Programming\WEEK 4-6\CCS711.pdf
2024-08-30 14:06:25 WAT	2024-08-30 14:06:25 WAT	GOJE ROOPRA			1.7	img_D1\1st_Semester\CCS712 Introduction to Cyber Security\INTRO TO CYBER SECURITY - CCS 712.pdf
2024-10-05 15:04:00 WAT	2024-10-05 15:32:00 WAT	ameerah07@gmail.com	Microsoft Office Word	ameerah07@gmail.com		img_D1\AL\TODAYAN.docx
2024-10-16 20:18:18 WAT	2024-10-25 10:11:03 WAT	Microsoft account			1.6	img_D1\1st_Semester\CCS 711 Introduction to Computer Programming\Group JAVA Programmes.pdf
2024-11-28 13:33:35 WAT	2024-11-08 13:33:35 WAT	ProGenius	Microsoft Office PowerPoint	ProGenius		img_D1\1st_Semester\CCS712 Introduction to Cyber Security\CyberThreats\CyberThreatAssignment.pptx
2024-11-28 09:47:00 WAT	2024-11-28 11:44:00 WAT	USER	Microsoft Office Word	nathanel nakotsu		img_D1\1st_Semester\CCS 710 Cyber Risk, Challenges and Mitigation\CCS 710_CYBER RISK MANAGEMENT AND MITIGATION
2024-11-28 09:47:00 WAT	2024-12-09 09:09:00 WAT	USER	Microsoft Office Word	nathanel nakotsu		img_D1\1st_Semester\CCS 710 Cyber Risk, Challenges and Mitigation\CCS 710_CYBER RISK MANAGEMENT AND MITIGATION
2024-12-10 10:44:00 WAT	2024-12-10 12:42:00 WAT	nathanel nakotsu	Microsoft Office Word	nathanel nakotsu		img_D1\1st_Semester\CCS 711 Introduction to Computer Programming\CCS 711 TEST and EXAMS Q&A.docx
2025-08-28 11:59:21 WAT	2025-08-28 11:59:21 WAT	Mr SAAD			1.5	img_D1\Introduction To Programming\Problem Solving.pdf
2025-08-28 21:32:00 WAT	2025-08-29 18:33:00 WAT	python.docx	Microsoft Office Word	Esther Nadele		img_D1\Footprinting_Movie_Report_Group1.docx
2025-10-09 17:05:28 WAT	2025-10-09 17:05:39 WAT		Microsoft Office PowerPoint			img_D1\The Science of Cyber-Risk.pptx

PREPARED BY GROUP 1 USING AUTOPSY 4.21.0 ON 26/10/2025

Activate Windows
Go to Settings to activate Windows.

